

Scalability & Future Plan

Date	15 March 2024
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current Credit Card Approval Prediction solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state, ensuring long-term viability and impact.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Model trained on a static historical dataset; does not retrain automatically as new applicant data comes in	Prediction accuracy may drift over time as economic conditions and applicant behavior change	High
2	Flask application currently handles a limited number of concurrent prediction requests	Performance may degrade if many analysts or customers submit requests simultaneously	Medium
3	Uses a fixed feature set (Gender, Income Type, Annual Income, Employment Duration, Education Level) without support for additional applicant attributes	Limits the model's ability to capture more nuanced risk factors, potentially affecting accuracy	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Single Flask instance serving one user/request at a time during local testing	Deploy behind a load balancer with multiple containerized Flask instances (Docker) to handle concurrent bank analysts and customers
2	Data Storage	Historical applicant data stored in flat CSV files used for model training	Migrate to a scalable cloud database (e.g., IBM Db2 or a managed SQL/NoSQL service) to support growing applicant records
3	Performance	Model inference runs synchronously within the Flask app on a single server	Leverage IBM Watson Machine Learning's auto-scaling cloud deployment for faster, parallelized real-time predictions
4	Security	Basic input validation on the web form with no authentication layer	Add role-based authentication (analyst, compliance officer, customer), HTTPS enforcement, and encrypted storage of sensitive applicant data

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Automated model retraining pipeline using newly collected applicant data	3-6 months	Keeps predictions accurate as applicant trends and economic conditions evolve

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Phase 3	Mobile-friendly interface and REST API for integration with existing bank systems	6-9 months	Wider accessibility for analysts and customers, easier integration with core banking software
Phase 4	Explainable AI (SHAP/LIME) dashboard to justify approval/rejection decisions to applicants and regulators	9-12 months	Improves transparency, regulatory compliance, and customer trust in automated decisions